English Dative Preferences as Statistical Inference over Verb-Specific Distributions

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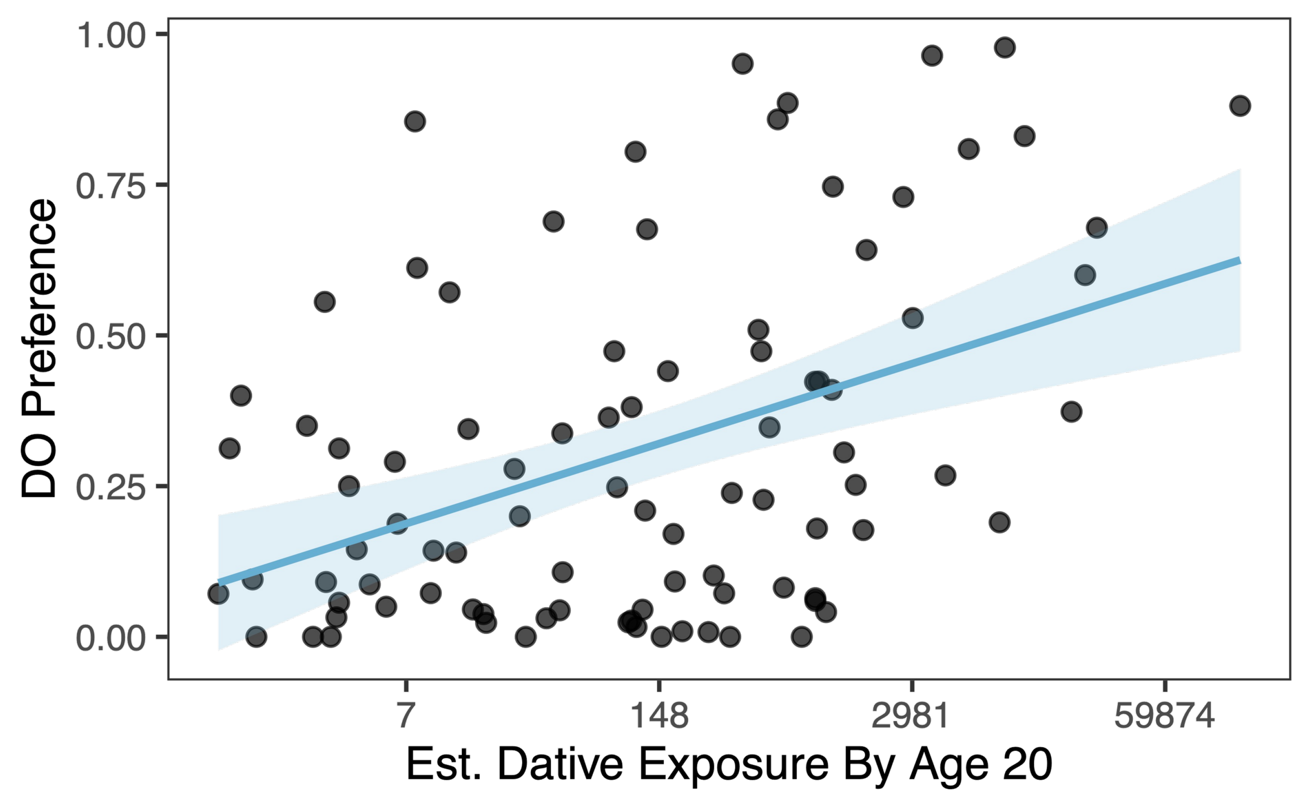
OVERVIEW

- More **frequent** English dative verbs show a stronger preference for the **double object (DO)** form (e.g., *give the kid a book*) over the **prepositional object (PO)** form (e.g., *give a book to the kid*) [1]
- Cumulative effects across generations from two sources:
  - **misattribution of abstract constraint effects** (e.g., given-before-new; [2]) to verb-specific biases
    - verb frequency ↑, contexts involving DO-favoring constraints ↑
  - **influence from non-dative occurrences** (often PO-patterning, e.g., *drive the kid to school*; [3,4])
    - verb frequency ↑, non-dative occurrences ↓

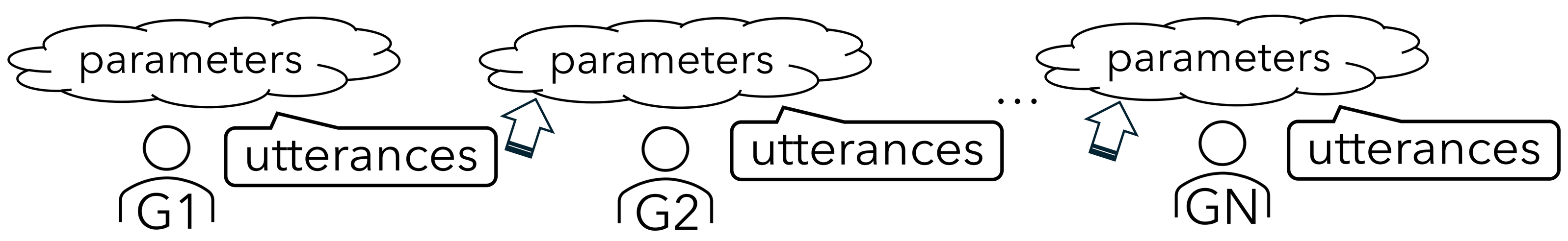
RQ: How does this pattern emerge?



Adapted from [1]

METHODS

- Iterated learning framework



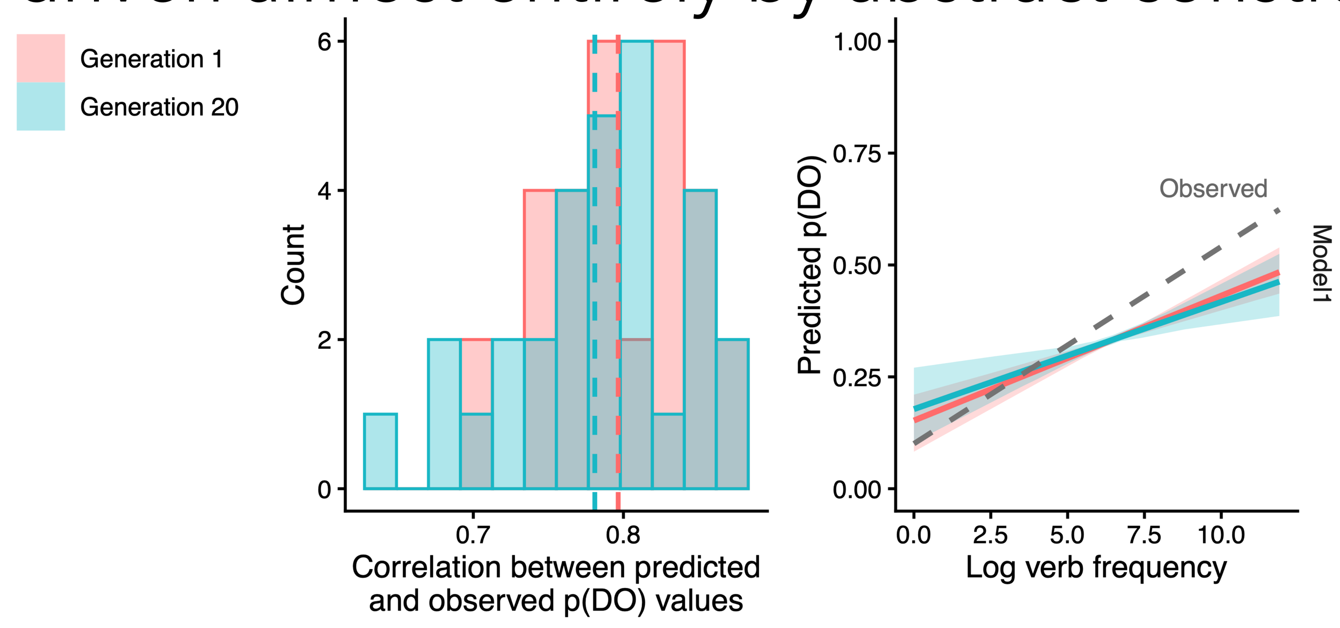
  - Each utterance token *i* (total # for each *v* = estimated exposure by 20 from [1]) is sampled as DO/PO with $p(DO)_{v,i}$ $= logit^{-1}$ (α baseline DO bias $+ u_v$ idiosyncratic DO bias $+ \eta_{v,i}$) abstract constraint effect for *i*
    - α and u_v: initialized around 0 for G1
    - $\eta_{v,i}$: sampled from an annotated corpus [1]
      DO, give, constraint1+, constraint2-, … $\eta_{give,1}$
      PO, give, constraint1-, constraint2+, … $\eta_{give,2}$
      … …
      y (DO/PO) ~ 1 + c1 + c2 + … + (1 | Verb)
  - Speaker in each G updates α and u_v based on a mixed-effects logistic regression fitted to the data
    - one desirable consequence: frequency-dependent distributional learning (u_v is estimated more strongly for higher frequency items) [4,5]
  - Four speaker types are modeled (2x2):

	$+\eta$	$-\eta$
−ND	Model 1	Model 2
+ND	Model 3	Model 4

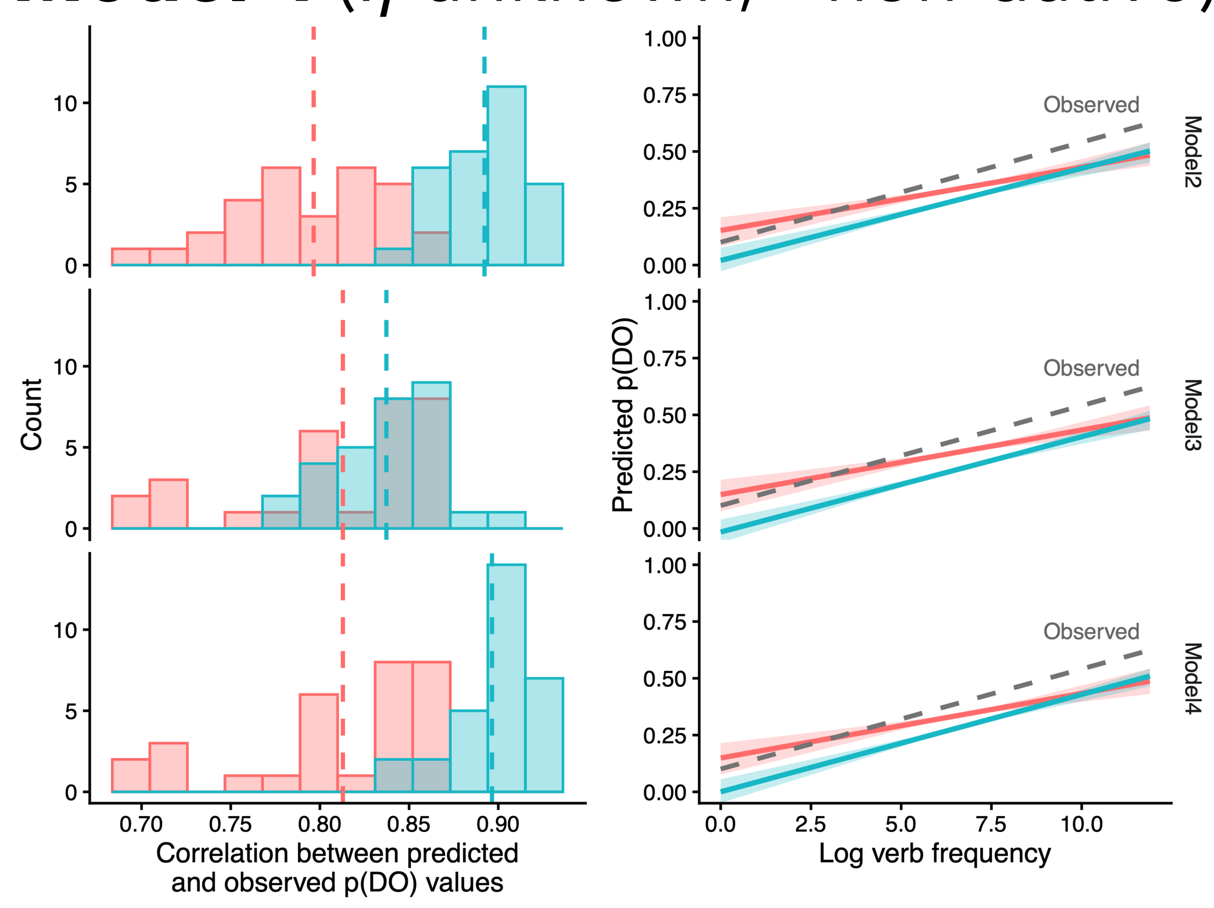
- **[± η]:** with or without knowledge of η for each token (i.e., fitting $p(DO)_{v,i} = logit^{-1}(\alpha + u_v + \eta_{v,i})$ or $p(DO)_{v,i} = logit^{-1}(\alpha + u_v)$)
- **[± ND]:** dative + non-dative input ($\eta = 0$; observations weighted 2:1; [4]) vs. dative input only

RESULTS

- 20 iterations, 30 chains with different random initializations
  - Key evaluation: improvement in alignment (from G1 to G20) between model outputs and corpus-observed values from [1]
- **Model 1** (η given, dative-only)
  - α and u_v remain close to 0; patterns are driven almost entirely by abstract constraints



- **Model 2** (η unknown, dative-only), **Model 3** (η given, +non-dative), & **Model 4** (η unknown, +non-dative)



DISCUSSION

English dative preference patterns may emerge through speakers' tracking of verb distributions (non-trivial α and u_v updates), specifically via **(1) misattribution** of abstract constraint effects to verb-specific preferences, and/or **(2) (mis-)learning** from non-dative input

Acknowledgment: This study was supported by NSF #2334255 to E.M. **References:** [1] Goodwin et al. (2025). *Proc. CogSci.* // [2] Bresnan et al. (2007). *Cognitive Foundations of Interpretation.* // [3] Bock & Loebell (1990). *Cognition.* // [4] Goodwin et al. (2026). HSP2026. // [5] Kapatsinski (2021). *Front. Psychol.*